

The total for HW10 is 72 points:

P1(17): (a) it is endothermic; temperature will go down; it takes more energy to break the bonds than the energy released from forming the bonds in the products; each of them worth 2 points. (b) The value of U worth 4 points; an explanation of the internal energy in physics worth 2 points. (c) The right value of Q worth 3 points; if you didn't specify the direction of Q , I assumed a positive sign means flowing into the reactor, which worth 2 points.

P2(15): Being able to derive the right chemical reaction and equations worth 8 points; the final right value of enthalpy worth 7 points.

P3(12): Reaching the right enthalpy change worth 6 points; mass or mole Fe worth another 6 points.

P4(16): each of the values worth 4 points.

P5(12): To be "as quantitative as possible", you need reach the enthalpy and temperature change respectively, and each of them worth 6 points.